Harvard University — Quantum Science & Engineering (PhD)

Statement of Purpose: Quantum Information Foundations, Circuit Complexity, and Error Mitigation

Candidate:	Sham Satish Thakare (PRN: 72159719L, Seat: B191102032)
Target Program:	PhD in Quantum Science and Engineering (QSE), Harvard University
Application Deadline:	December 15, 2026 (5:00 PM EST)
Transcript Distinction:	Course 417523A: Quantum AI — Grade O (Outstanding, 10.0/10.0)
Academic Distinctions:	Gold Medal in Mathematics (2024), 8.70 CGPA (First Class with Distinction), 9.53 Final SGPA (Rank 1)
Target Faculty:	Prof. Mikhail Lukin (Physics / QSE), Prof. Boaz Barak (SEAS / CS / QSE), Prof. Susanne Yelin (Physics / QSE)

Statement of Purpose

The realization of fault-tolerant quantum computation depends on solving a dual challenge: deriving fundamental theoretical limits on quantum information processing while engineering robust compilation algorithms that mitigate decoherence in physical hardware. Having earned the Gold Medal in Mathematics (2024) and attained the highest possible academic distinction (Grade O, 10.0/10.0 Outstanding) in university Quantum Artificial Intelligence (Course 417523A) at Savitribai Phule Pune University, I have built a rigorous theoretical and computational foundation in quantum information science. At Harvard's Quantum Science and Engineering (QSE) program, I seek to pursue doctoral research at the convergence of quantum complexity theory, optimal circuit synthesis, and stabilizer-based error mitigation architectures.

My academic preparation has focused on mastering the mathematical formalism of quantum states and operators. In my formal studies and peer mentorship, I have worked extensively with the multi-qubit density matrix formalism rho in S(H), satisfying rho >= 0 and Tr(rho) = 1, and quantified entanglement entropy via the von Neumann entropy S(rho) = -Tr(rho log rho). I explored the operational significance of quantum fidelity F(rho, sigma) in bounding state distinguishability under trace-preserving completely positive (CPTP) noise maps. Furthermore, I studied the complexity separations governing quantum advantage, examining the structural inclusions BPP $\subseteq$ BQP $\subseteq$ PSPACE and the query complexity speedups of the quantum search and phase estimation algorithms.

During my undergraduate studies and subsequent pre-doctoral research at Capital One Systems Research, I synthesized this theoretical background with algorithmic implementation. Serving as a Peer Academic Mentor in Discrete Mathematics and Quantum Information Foundations, I led student cohorts through rigorous proofs of the Gottesman-Knill theorem, stabilizer code projections (S_i |psi> = |psi>), and surface code syndrome extraction cycles. Computationally, I leveraged Qiskit and Cirq to model error-mitigation techniques—including zero-noise extrapolation (ZNE) and randomized benchmarking—and analyzed tensor network contractions (matrix product states and projected entangled pair states) for simulating shallow quantum circuits. This dual

fluency in abstract linear algebra and practical circuit synthesis confirmed my ambition to contribute to scalable quantum computing architectures.

Harvard's QSE program offers the premier interdisciplinary environment in the world for quantum science. I am deeply inspired by the groundbreaking experimental and theoretical achievements of Prof. Mikhail Lukin in neutral-atom quantum processors, quantum error correction with logical qubits, and programmable quantum simulators. I am keen to explore how algorithmic circuit compilation and fault-tolerant scheduling can be optimized for neutral-atom array topologies. I am equally eager to learn from Prof. Boaz Barak, exploring theoretical separations between classical and quantum computational complexity and the power of quantum interactive proofs. Additionally, the theoretical investigations of Prof. Susanne Yelin in cooperative quantum dynamics, quantum memory, and quantum information processing in many-body systems resonate strongly with my interests in modeling decoherence channels.

With a firm foundation in mathematical rigor, certified academic excellence in quantum computing, and a passion for foundational discovery, I am eager to dedicate my doctoral studies to advancing the theoretical and practical boundaries of quantum science at Harvard University.